

Saurabh Kumar

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EDUCATION

Dr. A.P.J. Abdul Kalam Technical University
Bachelor of Technology in Computer Science and Engineering

Uttar Pradesh, India
Aug. 2023 – June 2027

PROFESSIONAL SUMMARY

Computer Science student specializing in full-stack development and machine learning. Engineered 3 production applications using React, Node.js, MongoDB, and Python, supporting 500+ active users with 99% system reliability. Demonstrated expertise in AI integration, secure authentication, and cloud deployment. Passionate about leveraging technology to solve real-world problems in healthcare and education.

EXPERIENCE

Open Source Contributor Jan. 2024 – Present
GitHub Community *Remote*

- Contributing to React.js ecosystem projects by fixing bugs, improving documentation, and implementing new features.
- Collaborated with maintainers of popular npm packages to enhance functionality and resolve critical issues affecting 10,000+ users.
- Submitted pull requests to TensorFlow.js repository for improving model performance and adding new examples.
- Active participant in GitHub Discussions, providing technical support and code reviews for community members.
- Maintained personal open-source projects with 20+ repositories focusing on web development and ML tools.

PROJECTS

StudyMate - AI-Powered Learning Platform | *React, Node.js, MongoDB, Google Gemini AI* Jan. 2024

- Engineered full-stack AI study platform featuring quiz generation, intelligent tutoring chatbot, and document processing.
- Architected JWT-based authentication and theme management system serving 500+ concurrent users.
- Leveraged Google Gemini AI for automated quiz creation from PDF uploads achieving 95% accuracy.
- Deployed scalable infrastructure on Render with MongoDB Atlas maintaining 99% uptime.
- Links: [GitHub](#)

HavenTo - Accommodation Booking Platform | *React, Node.js, MongoDB, Express* Sep. 2023

- Developed production-grade booking platform with OTP email verification and session-based authentication.
- Established enterprise security infrastructure including DDoS protection, rate limiting, and Helmet.js headers.
- Designed image upload system using Multer with role-based access control for Guest and Host users.
- Launched on Vercel and Render with environment-specific configurations for production scalability.
- Links: [GitHub](#) | [Live Demo](#)

Brain Tumor Detection System | *Python, TensorFlow, Node.js, Express, React, MongoDB* Dec. 2023

- Architected medical imaging platform with hospital authentication using MongoDB Atlas and bcrypt encryption.
- Trained CNN model with VGG16 transfer learning achieving 78% accuracy and 97% tumor sensitivity on 253 scans.
- Integrated Python ML pipeline with Express.js API via child processes delivering predictions under 2 seconds.
- Established secure authentication framework with protected routes and CORS configuration.
- Designed React dashboard with modern UI for authorized MRI analysis and confidence visualization.
- Links: [GitHub](#)

TECHNICAL SKILLS

Languages: C, C++ (Data Structures and Algorithms), JavaScript, HTML/CSS, SQL, Python
Web Development: React.js, Node.js, Express.js, TailwindCSS, Vite, RESTful APIs
Databases: MongoDB, MySQL, MongoDB Atlas
Machine Learning: TensorFlow, Keras, NumPy, Pandas, Matplotlib, Seaborn, CNN, Transfer Learning
Security: JWT, bcrypt, DDoS Protection, XSS Prevention, Rate Limiting, Helmet.js, CORS
Developer Tools: Git, GitHub, VS Code, Postman, npm, Linux, Docker
Cloud and Deployment: Render, Vercel, MongoDB Atlas, CI/CD

ACHIEVEMENTS

Contributed to 5+ open-source projects including React.js ecosystem and TensorFlow.js with merged pull requests.
Developed and deployed 3 production-ready applications on cloud platforms serving 500+ total active users.
Achieved 95% accuracy in AI quiz generation system and 78% accuracy in medical tumor detection model.
Built scalable APIs handling 10,000+ daily requests with 99% uptime and comprehensive error handling.
Solved 100+ algorithmic problems on LeetCode and GeeksforGeeks demonstrating strong DSA proficiency.
Participated in college hackathons and coding competitions, securing recognition for innovative solutions.

CERTIFICATIONS

SimpliLearn: Frontend Development – Dec. 2023

INTERESTS

Interdisciplinary Studies: Neuroscience, Cellular Biology, Computational Biology